



The Oracle Asymmetry

Where evolutionary self-improvement of an agent gets measurable traction

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Organization: Agent Self-Evolution

Repository: github.com/jramos/agent-self-evolution

Abstract

A framework for evolving an agent's skills, tool descriptions, prompts, and code was exercised across many experiments. The consolidated result is an **asymmetry**. Self-evolution produced deploy-grade traction in exactly one regime: when a ground-truth **executable oracle** stands between the artifact and the verdict — code under deterministic tests, with no agent in the loop to absorb the change. On real historical bugs this re-derived a correct, oracle-matching fix for **12 of 20** bugs (deploy-reachable **0.60**, Wilson 95% CI [0.39, 0.78], pre-registered kill line 0.10, verdict **GREEN**). Where the fitness signal is instead an LLM judge or a capable agent's behavior, we measured **no detectable effect at our statistical power, on a single capable-agent class**, in both directions — improving the artifact did not help and degrading it did not hurt. That bounds the detectable effect; it is not a proof of inertness. The boundary is sharp: traction tracks how mechanical and agent-free the verdict is.

1. The positive result: executable-oracle code re-derivation

Real historical tool bugs were harvested from an active repository's git stream, each carrying its upstream fix commit as a **ground-truth oracle**. The loop repairs the buggy parent in an isolated worktree from failing-test feedback; a repair counts only if it passes the full fix-commit test set and behaviorally matches the upstream fix. The **organism** (one bug) is the unit of analysis, and a bug is **deploy-reachable** when a majority of independent seeds produce an oracle-matching fix.

Estimand (organism-level)	Value
Deploy-reachable	12 / 20 = 0.60
Wilson 95% CI	[0.387, 0.781]
Cluster bootstrap (mean, CI)	0.602, [0.40, 0.80]
P(below 0.10 kill line)	0.0
ICC / design effect / effective-N	0.326 / 1.65 / 36.3
Verdict / cost	GREEN / \$3.10

The honest unit is the organism. The pooled per-seed rate (41/60 = 0.68) is recorded only *for contrast* and labelled dishonest in the source artifact, because seed correlation (ICC 0.33) inflates its apparent precision — averaging seeds within a bug is pseudo-replication. An earlier, smaller pilot on an easier tool tier was consistent (deploy-reachable 0.75 [0.41, 0.93], smaller N); the N=20 campaign above is canonical because it is larger and pre-registered. Why this regime works: the test is deterministic and the verdict is mechanical, so nothing between the candidate and pass/fail can quietly compensate for a bad artifact — unlike a capable agent.

2. The null: capable-agent artifact decoupling

On a capable agent (gpt-5-mini class), artifact quality and correctness were **decoupled from behavior in both directions**:

- **Improving did not help.** Evolving skills, tool descriptions, and turn-level sequences produced null deploy results — including under a **judge-free test-command verdict**, so judge saturation is not the cause.
- **Degrading did not hurt.** A strongly misdirecting tool description, and even an **actively false** one ("merges and preserves content, never deletes"), each produced 0/7 caught-losses against a 0/7 A/A false-alarm floor. The agent routes from the tool name, the task, and its own prior — over the description text.

This is a real null, not a broken detector: a delivery **canary** (a description that changes which file is written) separates cleanly at 1.00 vs 0.00, so the rig can see a genuine behavior difference. There was simply nothing to catch. Stated honestly, this is a failure to reject the null on **one capable-agent class**, for tools whose behavior the agent can infer from their name. It bounds the detectable effect; it does not prove universal inertness. The coupling is expected to reappear where the agent cannot infer behavior from priors — a weaker tier, or a novel tool with a non-obvious contract — neither of which was a clean test here.

3. The dead ends, as boundary pilots

These are small-N negative pilots. They did not supply usable signal; they mark where we looked and why the positive regime is narrow — not proofs of impossibility.

- **Metamorphic verification (no oracle): 0/8.** Generic invariants (idempotence, absent-input no-op, parse-shape) caught 0 of 8 real bugs. Real tool bugs are input-specific edge cases that do not violate broad invariants.
- **Held-out-split independence: only 3/8.** For an independent anti-gaming gate a bug needs tests input-diverse enough to split; only 3 of 8 did. Real bugs are usually caught by the single test the fix authored, so the independent held-out product is deferred — the oracle sidesteps it.
- **Dependency-regression supply: 0.** Over a year of one active repo's history there were zero clean, tool-local dependency-version regressions: manifest-touching commits were broad migrations or feature additions.

4. The unifying thesis

Put the three classes together and the boundary is sharp. Self-evolution gets traction in proportion to **how mechanical and agent-free the verdict is**. A deterministic test on executed code (oracle present, no agent) yields a deploy-grade gradient. An LLM judge or a capable agent's behavior (no executable oracle; the agent absorbs the artifact) yields no detectable effect at our power. So the evolution and validation pipeline is high-value where an executable oracle exists — code, which the shipped propose-only triage sentinel feeds directly — and low-value for artifact quality on

capable agents. Remaining artifact-quality work presupposes a discriminating signal this campaign shows is absent on capable agents; pursue it only on a tier or surface where the coupling exists.

5. Limitations

- The null results bound an effect on **one capable-agent class**; a different model tier or a novel-contract tool could couple. Untested here, by design.
- The positive result is N=20 organisms on one repository's bug stream. It is GREEN with a pre-registered kill line, but it is one corpus.
- The dead-end pilots are small-N (8 organisms; one year of one repo). They steer the search; they are not impossibility proofs.